Supplementary Material

Per-instance Comparison with Published IJSP Solvers on All 70 Taillard Instances

Companion to: "Neighbourhood Structures and Ranking Operators

for the Interval Job Shop Scheduling Problem"

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Description

This document reports per-instance RE (%) and median runtime for all 70 Taillard Interval Job Shop Scheduling Problem (IJSP) instances, comparing four solvers: *fEABC* [1], *ESABC* [2], $TS\text{-}N_2$ (this work), and $TS\text{-}N_8$ (this work).

Metric. $\mathrm{RE}(\%) = (E[\mathbf{C}_{\max}] - \mathrm{LB}) / \mathrm{LB} \times 100$, where LB is the published lower bound of the *crisp* (deterministic) counterpart of each Taillard instance [3]. The interval IJSP and the crisp JSP are structurally distinct optimisation problems, so these crisp lower bounds are used here only as a common, hardware-independent reference yardstick; they are *indicative* for the interval setting and do not constitute a strict optimality gap.

Columns. For each solver: *Best* = best RE (%) over 30 runs; *Avg* = mean RE (%) over 30 runs (bold = row minimum); t = median runtime in seconds (TS) or average runtime in seconds (ABC methods).

Instances. The 70 Taillard IJSP instances (TA1–TA70) are derived from the classical Taillard JSP benchmarks [3] by applying a ±7.5% symmetric interval perturbation to each nominal processing time: $\mathbf{p}_{ij} = [0.925\, p^*_{ij},\ 1.075\, p^*_{ij}]$. Seven size classes are covered: 15×15 (TA1–10), 20×15 (TA11–20), 20×20 (TA21–30), 30×15 (TA31–40), 30×20 (TA41–50), 50×15 (TA51–60), 50×20 (TA61–70).

TS configuration. Irace-tuned tabu search (Phase B of the main paper), 30 independent runs per instance, 15-minute wall-clock time limit per run, on an Intel Xeon E5-2680 v4 at 2.40 GHz (WSL2 Linux).

Irace training instances. The irace tuning of Phase B (Section 8.1 of the main paper) used a stratified random subset of 20 of the 82 benchmark instances (≈ 24%), drawn uniformly at random within each size stratum. The 62 instances *not* included in the list below were held out from tuning and contribute the unbiased evaluation reported in the main paper and in Tables 1–2.

- *Classical* (3 of 12): `ft10`, `abz7`, `la27`.
- *Taillard* 15×15 (2 of 10): TA3, TA8.
- *Taillard* 20×15 (2 of 10): TA12, TA17.
- *Taillard* 20×20 (3 of 10): TA21, TA24, TA29.
- *Taillard* 30×15 (2 of 10): TA32, TA36.
- *Taillard* 30×20 (3 of 10): TA43, TA47, TA50.
- *Taillard* 50×15 (2 of 10): TA52, TA57.
- *Taillard* 50×20 (3 of 10): TA63, TA67, TA70.

The full irace input files (`instances.txt`, `parameters.txt`, `scenario.txt`) and the resulting `iracedump.rda` files for each of the five neighbourhoods are archived at [4].

Table 1: Taillard IJSP instances, Part I (TA1–TA40). RE (%) = $(E[\mathbf{C}_{\max}] - \mathrm{LB}) / \mathrm{LB} \times 100$. All methods: 30 runs. Bold: row-minimum average RE. t = median runtime (s) for TS; average for ABC methods.

			fEABC [1]			*ESABC* [2]			TS-N_2 (ours)			TS-N_8 (ours)		
Size	Inst.	LB	Best	Avg	t	Best	Avg	t	Best	Avg	t	Best	Avg	t
15×15	TA1	1231	4.71	6.73	3.7	4.51	7.27	5.4	1.26	**1.85**	20	1.26	2.09	34
	TA2	1244	2.65	4.20	3.9	3.26	4.87	5.2	0.24	1.65	19	0.88	**1.57**	37
	TA3	1218	3.65	5.63	4.4	3.74	5.57	5.7	1.35	2.34	23	1.35	**2.28**	35
	TA4	1175	4.38	6.99	4.0	5.15	6.96	5.9	0.34	1.31	21	0.34	**1.06**	37
	TA5	1224	3.19	4.89	3.5	2.49	4.27	5.0	1.06	2.22	19	1.06	**1.94**	33
	TA6	1238	3.39	5.61	4.2	2.34	4.89	5.5	1.29	1.97	20	1.62	**1.92**	33
	TA7	1227	2.81	4.86	3.6	2.57	4.42	5.3	0.61	**1.77**	16	0.61	1.79	27
	TA8	1217	2.55	5.41	3.9	1.97	5.68	5.2	1.15	2.37	17	1.15	**2.18**	31
	TA9	1274	4.47	8.25	4.1	5.46	8.26	5.7	2.35	**2.73**	17	2.35	2.74	30
	TA10	1241	3.14	6.16	3.8	3.71	6.44	5.3	0.24	2.23	19	0.24	**1.90**	31
20×15	TA11	1357	6.26	9.37	6.6	6.74	8.37	9.7	2.76	**4.11**	45	2.76	4.30	65
	TA12	1367	4.46	5.99	5.8	2.93	5.28	8.5	1.57	**2.21**	31	1.65	2.25	59
	TA13	1342	6.26	8.88	7.1	7.19	9.25	9.8	2.38	**3.91**	43	2.01	4.00	68
	TA14	1345	3.75	5.02	6.3	2.49	4.43	8.6	0.45	**1.82**	35	0.74	1.95	55
	TA15	1339	5.94	8.53	6.3	4.59	7.05	9.6	2.43	**3.96**	44	2.80	4.18	65
	TA16	1360	7.02	8.98	6.3	6.84	9.22	8.6	1.95	3.28	38	2.10	**3.23**	60
	TA17	1462	5.75	8.72	6.1	4.31	8.06	8.0	1.92	**3.14**	40	2.02	3.21	58
	TA18	1377	9.40	11.79	6.6	9.37	11.63	9.6	3.70	**5.23**	44	3.96	5.87	75
	TA19	1332	8.52	11.36	5.7	7.96	9.70	8.9	1.99	4.54	48	1.91	**4.38**	86
	TA20	1348	4.97	8.09	6.2	5.27	7.02	10.2	2.15	3.54	41	1.97	**3.51**	70
20×20	TA21	1642	6.52	8.07	6.9	5.42	7.47	11.2	2.38	**3.75**	60	2.74	3.90	97
	TA22	1561	8.65	10.53	8.1	8.23	9.77	11.8	4.16	**5.88**	59	4.80	6.00	91
	TA23	1518	10.01	11.56	8.0	9.49	11.23	11.8	4.45	**5.80**	49	4.41	6.26	85
	TA24	1644	5.14	7.47	7.8	5.29	6.91	11.0	1.43	2.86	58	1.34	**2.80**	101
	TA25	1558	9.27	10.97	7.8	8.06	11.22	11.2	5.10	**6.50**	59	5.30	6.81	97
	TA26	1591	10.50	13.05	8.4	9.99	12.93	11.7	6.66	**8.02**	54	6.03	8.05	89
	TA27	1652	8.57	10.42	8.3	7.45	9.82	11.6	3.93	5.08	58	3.63	**5.03**	95
	TA28	1603	6.46	8.21	7.5	6.24	7.52	10.9	2.18	**3.27**	60	2.25	3.38	99
	TA29	1583	7.99	10.21	7.6	7.64	9.63	12.0	4.36	5.10	50	4.14	**4.88**	82
	TA30	1528	9.75	12.22	7.7	10.01	12.04	10.6	5.99	7.13	70	5.82	**7.06**	102
30×15	TA31	1764	6.01	8.78	12.5	5.70	7.81	18.7	1.02	**1.97**	102	0.54	2.17	171
	TA32	1774	10.68	13.07	13.4	9.53	11.84	19.8	4.17	**5.32**	127	3.83	5.55	211
	TA33	1788	9.40	11.71	13.3	9.03	10.58	18.8	3.75	**4.61**	106	3.16	4.77	172
	TA34	1828	8.23	9.63	12.2	7.28	8.76	17.3	3.39	4.73	98	3.04	**4.71**	135
	TA35	2007	1.47	3.25	11.7	1.59	3.92	13.0	0.00	**0.26**	61	0.00	0.27	94
	TA36	1819	7.94	9.63	12.7	6.24	8.83	18.2	1.29	**2.82**	105	1.18	2.87	174
	TA37	1771	8.16	9.70	12.7	7.06	9.11	16.4	2.88	**3.85**	113	3.16	4.06	192
	TA38	1673	8.52	10.65	12.1	7.29	9.51	18.9	3.17	**4.35**	122	2.81	4.48	193
	TA39	1795	6.13	8.35	13.2	5.46	7.82	18.0	1.45	2.67	90	1.67	**2.60**	142
	TA40	1651	11.18	13.42	13.1	11.21	12.71	17.8	4.18	**6.17**	102	4.82	6.57	153

Table 2: Taillard IJSP instances, Part II (TA41–TA70). RE (%) = $(E[\mathbf{C}_{\max}] - \mathrm{LB}) / \mathrm{LB} \times 100$. All methods: 30 runs. Bold: row-minimum average RE. t = median runtime (s) for TS; average for ABC methods.

			fEABC [1]			*ESABC* [2]			*TS*-N_2 (ours)			*TS*-N_8 (ours)		
Size	Inst.	LB	Best	Avg	t	Best	Avg	t	Best	Avg	t	Best	Avg	t
30 × 20	TA41	1906	17.58	20.55	16.1	16.34	19.24	22.4	9.68	**12.12**	158	10.36	12.31	257
	TA42	1884	14.15	17.45	15.3	14.46	16.86	20.8	6.58	**8.36**	192	6.53	8.77	323
	TA43	1809	16.06	18.13	16.5	13.85	16.74	24.2	5.67	**8.72**	197	6.77	8.79	294
	TA44	1948	11.88	14.41	16.8	10.47	12.78	25.9	4.72	**6.86**	151	5.78	7.12	233
	TA45	1997	6.99	9.45	17.0	6.76	8.40	24.2	2.70	**4.17**	132	2.88	4.37	220
	TA46	1957	13.00	15.91	18.2	12.31	14.87	26.2	6.87	**8.69**	168	5.93	8.70	276
	TA47	1807	16.55	20.44	16.3	15.33	18.64	24.0	9.91	**11.53**	161	10.29	12.20	274
	TA48	1912	11.90	13.90	15.1	11.30	13.88	22.1	6.15	**7.87**	161	6.22	7.91	267
	TA49	1931	11.08	13.19	17.1	10.56	12.98	21.3	5.00	**6.49**	155	4.79	6.64	230
	TA50	1833	16.56	20.36	16.3	16.50	18.83	22.1	9.22	**11.43**	163	10.37	11.66	264
50 × 15	TA51	2760	5.51	7.20	27.3	3.73	5.76	40.6	0.00	**0.02**	204	0.00	0.09	291
	TA52	2756	4.03	5.93	27.5	2.87	4.47	43.2	0.00	0.10	212	0.00	**0.10**	272
	TA53	2717	3.66	5.40	27.5	2.82	4.04	37.9	0.04	**0.04**	162	0.04	0.05	233
	TA54	2839	0.63	2.96	24.3	0.33	1.75	34.3	0.00	**0.02**	108	0.00	0.04	158
	TA55	2679	6.01	8.44	29.3	5.88	7.52	39.2	0.34	**0.87**	244	0.34	0.88	407
	TA56	2781	3.25	5.04	23.7	3.00	4.57	36.9	0.11	**0.27**	156	0.11	0.33	258
	TA57	2943	1.63	3.61	26.9	1.22	2.48	41.1	0.00	0.03	127	0.00	**0.00**	197
	TA58	2885	3.08	4.94	26.7	1.94	3.60	38.4	0.69	**0.69**	125	0.69	0.69	184
	TA59	2655	5.63	7.50	27.8	4.73	6.62	38.9	0.45	**0.75**	209	0.45	0.81	385
	TA60	2723	4.43	6.07	24.7	3.64	4.67	34.9	0.11	0.41	185	0.11	**0.35**	261
50 × 20	TA61	2868	6.76	8.84	33.5	5.81	7.77	49.2	0.47	**1.83**	542	0.47	2.18	784
	TA62	2869	9.41	11.74	33.7	8.66	10.40	44.7	3.66	**5.02**	460	3.31	5.08	653
	TA63	2755	7.64	9.19	30.6	6.35	7.87	45.8	0.45	1.69	545	0.42	**1.66**	750
	TA64	2702	6.94	8.36	33.6	5.94	6.78	47.0	0.59	1.50	521	0.13	**1.44**	689
	TA65	2725	8.42	9.98	33.8	7.30	9.05	47.5	1.19	**2.12**	574	1.08	2.32	741
	TA66	2845	6.80	8.56	32.1	6.05	7.55	42.8	1.20	2.05	527	0.93	**1.95**	655
	TA67	2825	6.50	8.95	34.0	6.50	8.24	45.4	1.08	2.30	585	1.31	**2.24**	788
	TA68	2784	6.66	8.01	32.7	4.87	6.85	46.6	0.29	**1.10**	421	0.27	1.15	534
	TA69	3071	5.08	6.59	29.6	3.70	5.90	45.0	0.00	**0.12**	440	0.00	0.38	461
	TA70	2995	8.43	9.91	30.9	7.20	8.72	42.9	1.59	**2.53**	664	1.80	2.90	701

References

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